

CREDPULSE

AI-Powered Credit Risk Intelligence Platform

End-to-End Enterprise Underwriting, TreeSHAP Explainability, Business Rules & NL-to-SQL Analytics

Candidate Assignment Submission • AI Engineer Role

Executive Summary & Objectives

Bridging predictive machine learning with conversational data exploration and audited governance

- **Business Context:**

Banks require fast, accurate, and regulatory-defensible loan underwriting decisions.

- **Core Challenge:**

Significant class imbalance (8.07% default rate) across 307K+ complex multi-table records.

- **Target Capabilities:**

Automated calibrated scoring, exact local explainability, policy guardrails, and natural language analytics.

Platform Pillars

- ✓ Predictive ML (LightGBM)
- ✓ Isotonic Calibration
- ✓ TreeSHAP Additivity
- ✓ Evidence-Based Rules
- ✓ Talk-to-Data NL-to-SQL
- ✓ React + Docker Ready

System Architecture & Layer Separation

Strict 6-layer modular design ensuring reproducibility and security

- **Layer 1: Ingestion & EDA:**

Optimized chunked loaders across 10 tables; automated quality & anomaly audit.

- **Layer 2: Feature Engineering:**

Domain financial ratios (DTI, Annuity/Income), bureau aggregates, leakage-safe CV.

- **Layer 3: Machine Learning:**

Tuned LightGBM + Isotonic calibration with F1-optimal risk thresholding.

- **Layer 4: Explainable AI:**

Exact polynomial-time TreeSHAP attributions with Adverse Action text generator.

- **Layer 5: Business Rules:**

6 empirical guardrails combining statistical thresholds and underwriting policy.

- **Layer 6: Talk-to-Data:**

SQLite analytical database (224MB) + Groq LLM & deterministic fallback.

Machine Learning Experiments & Benchmarks

5-Fold Stratified Cross-Validation with leakage-safe pipeline fitting

- **Model Selection:**

Tuned LightGBM achieved 0.7812 ROC-AUC and 0.2741 PR-AUC, outperforming all benchmarks.

- **Imbalance Strategy:**

Employed scale_pos_weight=11.4 to penalize minority-class default

- **Inference Latency:**

Sub-50ms execution per applicant, meeting strict production SLA req

Model	ROC-AUC	PR-AUC	F1	Brier
Logistic Reg.	0.7642	0.2443	0.3109	0.1968
Default LGBM	0.7733	0.2613	0.3236	0.1531
Tuned LGBM	0.7812	0.2741	0.3382	0.1482
XGBoost	0.7765	0.2678	0.3294	0.1505

Probability Calibration & Risk Bands

Converting raw classifier outputs into reliable, empirical default probabilities

- **Isotonic Calibration:**

`CalibratedClassifierCV(method='isotonic')` aligns predicted scores with true population rates.

- **Brier Score Reduction:**

Substantially reduced Brier score loss from 0.1531 to 0.1482.

- **Risk Tier LOW ($PD < 5\%$):**

Represents ~42% of applicants; observed default rate is only 2.4% (auto-approve cohort).

- **Risk Tier MEDIUM ($5\% \leq PD < 20\%$):**

Represents ~47% of applicants; observed default rate is 9.1% (standard underwriting).

- **Risk Tier HIGH ($PD \geq 20\%$):**

Represents ~11% of applicants; observed default rate is 24.8% (>2.5x base rate).

Explainable AI (TreeSHAP) & Adverse Action

Ensuring FCRA & ECOA regulatory defensibility via exact Shapley values

- **TreeSHAP Guarantee:**

Polynomial-time exact computation satisfying strict efficiency and additivity properties.

- **Top Risk Drivers:**

External bureau composites (EXT_SOURCE), Debt-to-Income, and historical late payment counts.

- **Protective Factors:**

Higher education status, mature age, established employment duration, and active collateral.

- **Plain-English Generation:**

Translates positive and negative SHAP attributions into clear, human-readable reasons.

Evidence-Based Business Rules Engine

Audited governance guardrails derived directly from empirical EDA and model behavior

- **BR-01: External Credit Scores:**

Mean EXT_SOURCE < 0.35 triggers high-risk warning (21.4% default cohort).

- **BR-02: High Debt Burden:**

Credit-to-Income > 4.0 triggers hard loan cap recommendation.

- **BR-03: Young Untested Borrower:**

Age < 25 with low credit score triggers manual underwriting verification.

- **BR-04: Prior Loan Refusals:**

>= 2 historical application rejections triggers elevated risk flag.

- **BR-05: Payment Delays:**

> 20% delayed payments triggers cash-flow stress warning.

- **BR-06: Prime Borrower Pass:**

High credit score + low debt burden fast-tracks auto-approval.

Talk-to-Data (NL-to-SQL) Intelligence

Conversational data exploration with zero hallucination and strict security

- **Dual Execution Engine:**

Groq Llama-3.3-70B cloud LLM + deterministic offline pattern matcher.

- **Zero Hallucination:**

Strict schema-only prompt context; all metrics generated from actual executed SQL.

- **Security Firewall:**

Enforces read-only SELECT; strictly blocks DROP, DELETE, INSERT, UPDATE, and ALTER.

- **Analytical SQLite DB:**

224MB database containing 307K applicants, bureau summaries, and repayment tables.

Full-Stack Web Interface & Containerization

State-of-the-art developer and user experience

- **React 19 + TailwindCSS v4:**

Dark-theme glassmorphism UI featuring interactive Recharts analytics and Framer Motion.

- **FastAPI Backend:**

High-performance asynchronous API serving ML inference, SHAP, Rules, and static React SPA.

- **Multi-Stage Dockerfile:**

Stage 1 builds frontend bundle; Stage 2 packages Python 3.11 runtime (<400MB).

- **One-Command Deployment:**

`docker-compose up --build` launches the complete platform at port 8000.